

1. Introduction to Generative AI

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Class Rep Recruitment

► <https://vimeo.com/316631158>

Which AI product have you used in the last week?

AI Evolution: Rule-based AI

- ▶ Approximate period: 1956 - 1980s.
- ▶ 1956, Dartmouth Workshop.
- ▶ John McCarthy coined the term "Artificial Intelligence".
- ▶ Famous proposal: "Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it."
- ▶ AI can learn human knowledge, follow rules defined by human experts.
- ▶ Example: Expert System.
- ▶ Subset of Symbolic AI.

First AI Winter: 1973 - 1979

- ▶ Early successes during 1960s: ELIZA (NLP)
- ▶ Promises were made; AI progress was little.
- ▶ "Machines will be capable, within twenty years, of doing any work a man can do." Herbert Simon, 1965.
- ▶ Book *Perceptrons* proves that a single-layer perceptron cannot even solve XOR problems.
- ▶ The Lighthill report concluded that AI failed to meet promises, 1973.
- ▶ UK stopped funding AI research after the report.
- ▶ Causes: limited computing power, little data, non-scalable rules.

Machine Learning: 1980s - Present

- ▶ Knowledge learned from data, not explicitly programmed by humans.
- ▶ Decision Tree CART, 1984.
- ▶ Bayesian Networks, 1985.
- ▶ Backpropagation, proposed 1974, popularised 1986.
- ▶ Supporting Vector Machine with kernel tricks, 1992.

Second AI Winter: mid 1980s - early 1990s

- ▶ Many Expert System Failures.
- ▶ AI was considered pseudoscience.
- ▶ Knowledge acquisition bottleneck.
- ▶ Rules became unmanageable.
- ▶ No learning ability.
- ▶ Expectations become unrealistic.

Deep Learning: 2012 - Present

- ▶ ImageNet Competition: 2010 - 2017.
- ▶ ImageNet: 14 million images, 20,000 categories.
- ▶ Use a subset, predict the class of a given image out of 1000.
- ▶ In 2012, AlexNet won the competition by a large margin.
- ▶ Previously, 26% error. AlexNet reduced it to 15%.
- ▶ DNN trained on large datasets using GPUs could outperform conventional ML methods.

Generative AI: 2013/2014 - Present

- ▶ Variational Autoencoder (VAE), 2013.
- ▶ Generative Adversarial Networks (GAN), 2014.
- ▶ Transformer, 2017.
- ▶ GPT-1, 2018.
- ▶ Stable Diffusion, 2022.
- ▶ ChatGPT, 2022.

What is Generative AI?

- ▶ Learns the underlying data distribution.
- ▶ Produces new samples that resemble the training data.
- ▶ "Generative artificial intelligence (GenAI) is the class of AI models that emulate the structure and characteristics of input data in order to generate derived synthetic content. This can include images, videos, audio, text, and other digital content.", NIST.